

Machine Learning (CAI-262)

Objectives:

- **Types of Clustering**
 - **K-means Clustering**
 - **Elbow method**
 - **Silhouette Score**
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Clustering

Clustering is an unsupervised machine-learning technique that involves organizing similar data points based on certain similarities or patterns in the data into groups or segments called **clusters**. Clustering is an unsupervised learning technique, which means that it does not require any labelled data to be trained. Clustering has many applications in various fields, including data analysis, image processing, pattern recognition, customer segmentation, and many others. Clustering can also help identify meaningful patterns or structures in large datasets, and it can also be used for data compression or simplification.

Types of Clustering

The clustering algorithms can be broadly classified into four categories

1. Centroid-based
2. Hierarchical
3. Distribution-based
4. Density-based

1. Centroid-based Clustering

Centroid-based clustering is a common method for grouping data points into clusters. The idea behind this method is to find a representative point or "centroid" for each cluster, which serves as the centre of the cluster. The centroid is typically defined as the mean or median of the data points in the cluster. The most popular centroid-based clustering algorithm is K-means, K-means has several advantages, including its simplicity, scalability, and efficiency. However, it also has some limitations, such as its sensitivity to initial centroid placement and the assumption that clusters are spherical and equally sized. To overcome these limitations, other algorithms like hierarchical clustering, DBSCAN, and Gaussian Mixture Models (GMMs) have been developed.

2. Hierarchical Clustering

Hierarchical clustering is a type of unsupervised machine-learning technique that aims to group similar data points into clusters based on their pairwise similarities. The key idea behind hierarchical clustering is to build a hierarchy of clusters, where each cluster is represented by a single data point, and the clusters at each level of the hierarchy are nested within the clusters at the next higher level. Hierarchical clustering can be performed using two main approaches: **agglomerative** and **divisive**. Agglomerative clustering starts with each data point as a separate cluster and merges them iteratively until all data points belong to a single cluster. Divisive clustering, on the other hand, starts with all data points in a single cluster and splits them recursively until each data point is in its own cluster. Hierarchical clustering can also be further divided into two subtypes: **single linkage** and **complete linkage**. In a single linkage, the distance between two clusters is defined as the shortest distance between any two points in the two clusters. In complete linkage, the distance between two clusters is defined as the maximum distance between any two points in the two clusters. Hierarchical clustering is a useful technique for exploring and visualizing high-dimensional data, as it can reveal natural groupings and structure in the data. However, it can be computationally expensive and sensitive to outliers and noise in the data.

3. Density-based Clustering

Density-based methods are a type of clustering algorithm used in unsupervised machine learning to group similar data points based on their density in a given feature space. These methods identify clusters by grouping together data points that are closely packed together and have a high density in the feature space while separating them from regions of lower density. The most popular density-based clustering algorithm is DBSCAN (Density-Based Spatial Clustering of Applications with Noise), which works by defining a neighbourhood around each data point and then expanding the cluster by adding neighbouring points that meet certain density criteria. Points that do not meet these criteria are considered outliers or noise. Another example of a density-based clustering algorithm is OPTICS (Ordering Points To Identify the Clustering Structure), which is similar to DBSCAN but provides a more flexible way of exploring the clustering structure of the data by generating a hierarchical clustering tree. Density-based methods are particularly useful for identifying clusters of arbitrary shape and size, as they do not make assumptions about the shape of the clusters or their number. They also have the advantage of being able to handle noisy data and outliers, which can be difficult for other clustering algorithms.

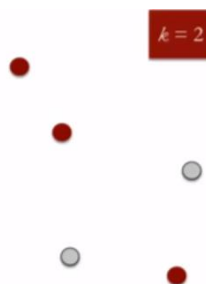
4. Distribution-based Clustering

Distribution-based clustering is a type of clustering algorithm that groups data points based on their probability distribution. This approach assumes that the data points are generated from a specific probability distribution, and aims to identify clusters that correspond to the different modes or components of this distribution. One common approach for distribution-based clustering is Gaussian Mixture Models (GMMs), which assume that the data points are generated from a mixture of several Gaussian distributions. The algorithm estimates the parameters of these Gaussian distributions, such as the means and covariances, and uses them to determine the probability of each data point belonging to each cluster. The clusters are then formed based on the highest probabilities. Another example of distribution-based clustering is the Dirichlet Process Mixture Model (DPMM), which is a non-parametric Bayesian approach. The DPMM assumes that the data points are generated from an infinite mixture of Gaussian distributions and uses a hierarchical Bayesian model to estimate the number of clusters and their parameters. Overall, distribution-based clustering can be a powerful tool for identifying complex patterns in data that may not be easily detectable using other clustering algorithms. However, it requires assumptions about the underlying probability distribution, and may not perform well if these assumptions are not met.

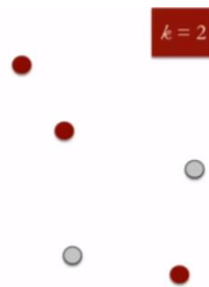
K-means Clustering:

K means is an iterative clustering algorithm that aims to find local maxima in each iteration. This algorithm works in these 5 steps:

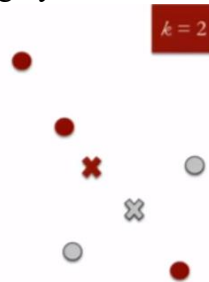
Step1: Specify the desired number of clusters K: Let us choose $k=2$ for these 5 data points in 2-D space.



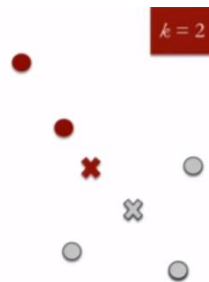
Step2: Randomly assign each data point to a cluster : Let's assign three points in cluster 1 shown using red colour and two points in cluster 2 shown using grey colour.



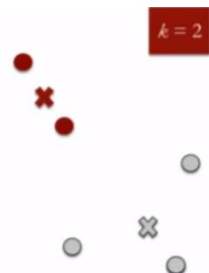
Step3: Compute cluster centroids: The centroid of data points in the red cluster is shown using red cross and those in grey cluster using grey cross.



Step4: Re-assign each point to the closest cluster centroid: Note that only the data point at the bottom is assigned to the red cluster even though it's closer to the centroid of grey cluster. Thus, we assign that data point into grey cluster.



Step5: Now, re-computing the centroids for both the clusters. Repeat until there will be no further switching of data points between two clusters.



Example1:

Consider the following set of points

P1(1,3), P2(2,2), P3(3,1), P4(4,2), P5(8,1), P6(8,5), P7(9,2), P8(10,1)

Consider that clusters are initialized at : C1(2,1) and C2(9,2)

Use Manhattan distance formula.

$$P(x, y) = |x_2 - x_1| + |y_2 - y_1|$$

- (a) Draw a table in the format below showing which point belongs to which cluster (clearly show distances for each point from both clusters)?

	C₁	C₂
P_i	Distance of P _i from C ₁	Distance of P _i from C ₂
P_j	Distance of P _j from C ₁	Distance of P _j from C ₂

SOLUTION

Calculate distance using given formula as follows and fill table

$$d(P_1, C_1) = |3 - 1| + |1 - 2| = |2| + |1| = 3$$

$$d(P_1, C_2) = |9 - 1| + |2 - 3| = |8| + |1| = 9$$

$$d(P_2, C_1) = |2 - 2| + |2 - 1| = |0| + |1| = 1$$

$$d(P_2, C_2) = |9 - 2| + |2 - 2| = |7| + |0| = 7$$

Points	C1	C2
P1	3	9
P2	1	7
P3	1	7
P4	3	5
P5	6	2
P6	10	4
P7	8	0
P8	8	2

Now we can decide which points belong to which cluster based on distance

Cluster1: P1, P2, P3, P4

Cluster2: P5, P6, P7, P8

- (b) Show the first iteration where the centroid centre shift from their default position C1(2,1) and C2(9,2) according to points calculated in part (a).

SOLUTION

Centroid1:

$$X = (1+2+3+4)/4$$

$$Y = (3+2+1+2)/4$$

New Centroid Centre: (2.5, 2)

Centroid2:

$$X = (8+8+9+10)/4$$

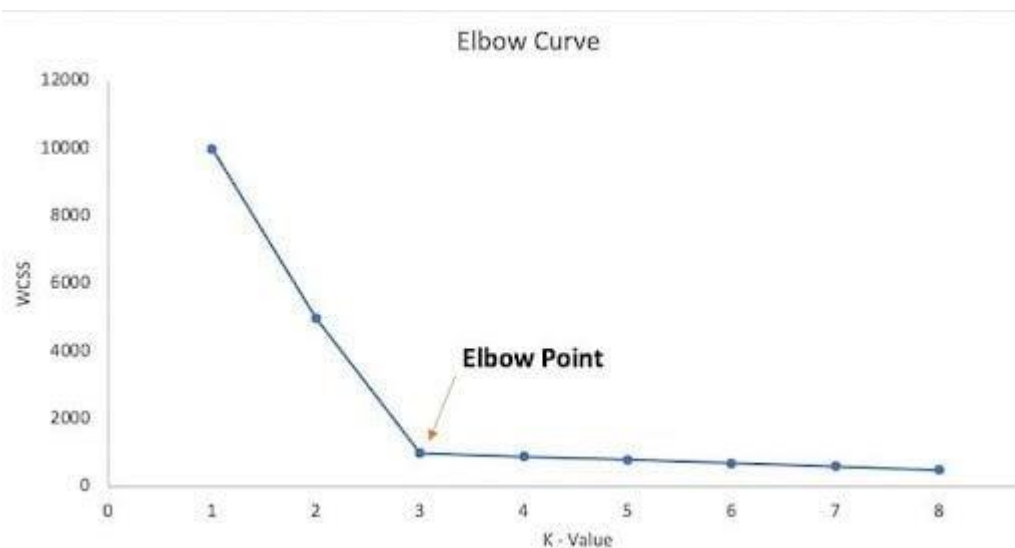
$$Y = (1+5+2+1)/4$$

New Centroid Centre: (8.75, 2.25)

Elbow Method

The elbow method is used to determine the optimal number of clusters in data. The method gets its name from the shape of the plot that is used to visualize the results, which looks like an elbow. To apply the elbow method, you first run a clustering algorithm, such as K-means, on the dataset for different values of K, where K represents the number of clusters. Then, for each value of K, you calculate the sum of squared distances between each data point and its nearest cluster centroid. This is known as the Within Cluster Sum of Squares (WCSS) also known as

inertia. Next, you plot the values of WCSS for each value of K. The resulting plot should show a downward trend in the WCSS as the number of clusters increases because the data points are divided into smaller and more homogeneous groups. However, at some point, the marginal decrease in WCSS starts to decrease as the number of clusters continues to increase, and this is the point where the elbow occurs in the plot. The optimal number of clusters is the value of K at which the marginal decrease in WCSS becomes small, and the elbow in the plot represents this point. The elbow method suggests that the number of clusters should be chosen at this point because this is where the additional clusters do not add significant information to the clustering analysis.



Performance Evaluation of Clusters

The **silhouette coefficient** is a metric that measures how well each data point fits into its assigned cluster. It combines information about both the **cohesion** (how close a data point is to other points in its own cluster) and the **separation** (how far a data point is from points in other clusters) of the data point.

The coefficient ranges from -1 to 1, where a value close to 1 indicates a well-clustered data point, a value close to 0 suggests overlapping clusters, and a value close to -1 indicates a misclassified data point.

A higher silhouette score indicates that the data points are well-clustered, with clear separation between clusters and tight cohesion within each cluster. Conversely, a lower silhouette score suggests that the clustering may be less accurate, with overlapping clusters or points that are not well-assigned to their respective clusters.

Calculating the Silhouette Coefficient: Step-by-Step

1. For each data point, calculate two values:

- Average distance to all other data points within the same cluster (cohesion).
 - Average distance to all data points in the nearest neighbouring cluster (separation).
2. Compute the silhouette coefficient for each data point using the formula:
- $$\text{silhouette coefficient} = (\text{separation} - \text{cohesion}) / \max(\text{separation}, \text{cohesion})$$
3. Calculate the average silhouette coefficient across all data points to obtain the overall silhouette score for the clustering result.

Example:

Let's consider 3 clusters, where each cluster has 3 (2-dimensional) points.

Cluster 1:

Point A1: (2, 5)

Point A2: (3, 4)

Point A3: (4, 6)

Cluster 2:

Point B1: (8, 3)

Point B2: (9, 2)

Point B3: (10, 5)

Cluster 3:

Point C1: (6, 10)

Point C2: (7, 8)

Point C3: (8, 9)

Step 1: Calculate Cohesion for Point A1

- Distance from Point A1 to Point A2: $\sqrt{(2-3)^2 + (5-4)^2} = \sqrt{2}$
- Distance from Point A1 to Point A3: $\sqrt{(2-4)^2 + (5-6)^2} = \sqrt{5}$
- Average cohesion for Point A1 = $(\sqrt{2} + \sqrt{5}) / 2 \approx 1.825$

Step 2: Calculate Separation for Point A1

- Distance from Point A1 to Point C1: $\sqrt{(2-6)^2 + (5-10)^2} = \sqrt{26}$
- Distance from Point A1 to Point C2: $\sqrt{(2-7)^2 + (5-8)^2} = \sqrt{18}$
- Distance from Point A1 to Point C3: $\sqrt{(2-8)^2 + (5-9)^2} = \sqrt{32}$
- Average separation for Point A1 = $(\sqrt{26} + \sqrt{18} + \sqrt{32}) / 3 \approx 5.333$

Step 3: Calculate Silhouette Coefficient for Point A1

- silhouette coefficient = $(\text{separation} - \text{cohesion}) / \max(\text{separation}, \text{cohesion})$
- silhouette coefficient = $(5.333 - 1.825) / \max(5.333, 1.825) \approx 0.657$

Step 4: Average Silhouette Coefficient

Calculate the average silhouette coefficient for all data points to obtain the overall silhouette score for the clustering result.